

5.2.4 Multiscaler (MCS)

MCS mode is demonstrated with the program `sis3820_mcs.c`. With respect to the fact that SDRAM mode with a fixed number of acquisitions is used, its kind of a minimum approach that does not make use of the full capabilities of the SIS3820.

The example uses the internal 10 MHz LNE source, which is run through the LNE prescaler to minimize the need for external signals. The number of acquisitions is preset with the acquisition preset register. The acquisition count register is used to determine completion of the acquisition process.

Find below the output of the program for 11 LNEs (corresponding to 10 time slices) with an input rate of 11 MHz moved across scaler channel 1 to channel 4 as the MCS acquisition is going on. Note that the CLR LED is pulsed with every LNE and that the S LED is on from the enable command until the acquisition preset has been reached.

```
mki@mki:~/sis1100/sis3820> ./sis3820_mcs
MCS mode, scan 1 completed
MCS mode, scan 2 completed
MCS mode, scan 3 completed
MCS mode, scan 4 completed
MCS mode, scan 5 completed
MCS mode, scan 6 completed
MCS mode, scan 7 completed
MCS mode, scan 8 completed
MCS mode, scan 9 completed
MCS mode, scan 10 completed
gotwords 40
scaler data
scan: 001 ch01 00a7d8f7 ch02 00000000 ch03 00000000 ch04 00000000
scan: 002 ch01 00a7d8f7 ch02 00000000 ch03 00000000 ch04 00000000
scan: 003 ch01 008f5010 ch02 00000000 ch03 00000000 ch04 00000000
scan: 004 ch01 00000000 ch02 003d3c9f ch03 00000000 ch04 00000000
scan: 005 ch01 00000000 ch02 00a7d8f7 ch03 00000000 ch04 00000000
scan: 006 ch01 00000000 ch02 002255a6 ch03 00000000 ch04 00000000
scan: 007 ch01 00000000 ch02 00000000 ch03 006ee942 ch04 00000000
scan: 008 ch01 00000000 ch02 00000000 ch03 0076e9f4 ch04 00000000
scan: 009 ch01 00000000 ch02 00000000 ch03 00000000 ch04 00438a6e
scan: 010 ch01 00000000 ch02 00000000 ch03 00000000 ch04 00a7d8f7
```